

ProtPure

BUILDING INDIA'S INDIGENOUS CHROMATOGRAPHY RESIN PLATFORM

Supporting Biopharma, Diagnostics,
Vaccines & Advanced Biotechnology

PROTPURE TECH PVT. LTD.



Enabling Affordable, Reliable &
Scalable Protein Purification Solutions



First Indian
Manufacturer of
Ni-NTA Agarose



Indigenous
Agarose-Based
Chromatography
Media



Affinity & Ion
Exchange
Chromatography
Solutions

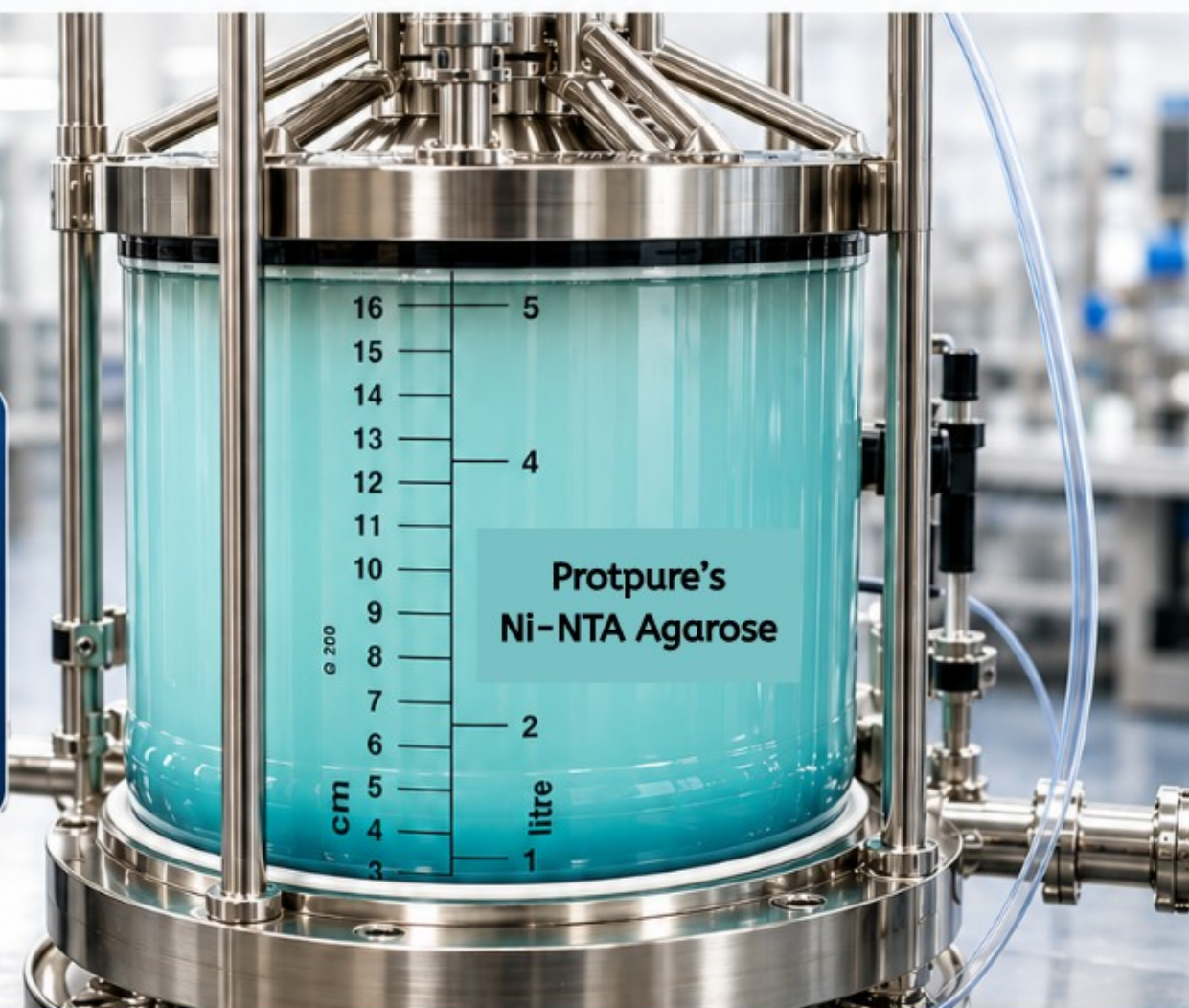


Supporting India's
Biotechnology &
Biopharma
Ecosystem



**Commercial
deployment**

of Protpure
chromatography
media



MADE IN INDIA

DEVELOPED IN INDIA

MANUFACTURED IN INDIA

PROTPURE AGAROSE SEC RESIN

Two Powerful Applications. One Reliable Resin.

1 SIZE EXCLUSION CHROMATOGRAPHY (SEC)

High-Resolution Molecular Weight Separation

KEY BENEFITS



High Resolution
Excellent separation with minimal peak overlap



Accurate MW Estimation
Excellent linearity ($R^2 = 0.9888$)



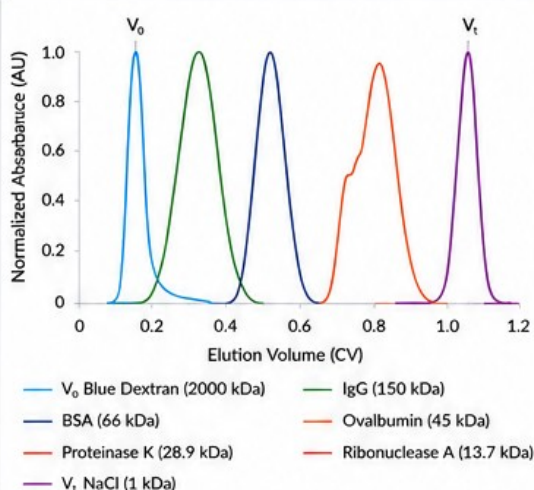
High Efficiency
1838.25 million plates with low HETP (0.0544 m)



Reproducible & Reliable
Consistent performance run after run

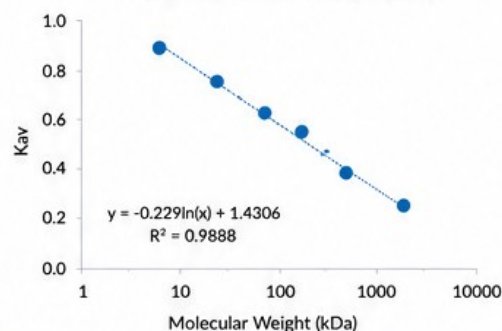
Protpure Agarose SEC Resin delivers high-resolution, size-dependent separation across a wide molecular weight range. Ideal for molecular weight estimation, aggregate analysis and purity assessment.

SEC SEPARATION PROFILE



SEC CALIBRATION

Kav vs Molecular Weight (kDa)



SEC CALIBRATION DATA

Protein / Standard	M.W (kDa)	CV Peak	Kav
Blue Dextran	2000	0.22	—
Immunoglobulin (IgG)	150	0.46	0.33
BSA	66	0.53	0.43
Ovalbumin	45	0.63	0.57
Proteinase K	28.9	0.80	0.81
Ribonuclease A	13.7	0.83	0.85
NaCl	1	0.94	—

COLUMN PERFORMANCE

Column	Column Diameter (mm)	Column Height (cm)	Column Volume (mL)	HETP (m)	No. of Plates (millions)
SEC Resin	50/600	40	785	0.0544	1838.25

2 DESALTING & BUFFER EXCHANGE

Efficient Protein-Salt Separation

Protpure Cross-linked Agarose Resin efficiently separates proteins from low molecular weight salts, enabling rapid desalting and buffer exchange with excellent reproducibility.



Protein First
BSA elutes before salt



Clear Separation
UV280 for protein and conductivity for salt

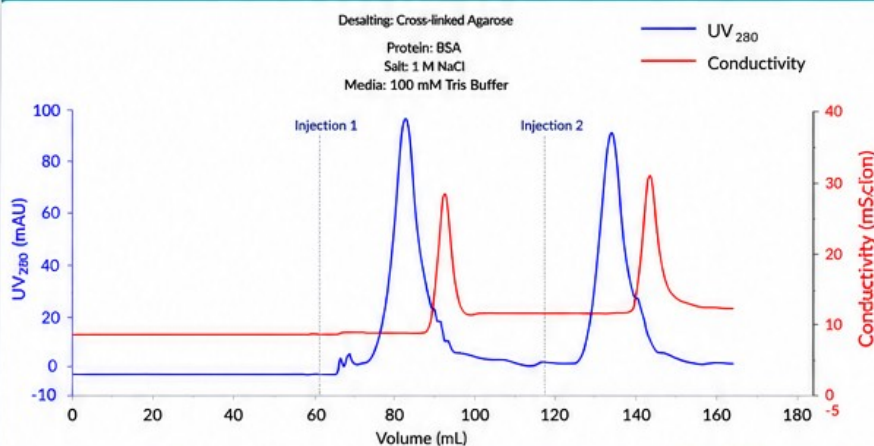


Reproducible Performance
Consistent separation across runs

COLUMN CONDITIONS (Desalting)

Column	Column Diameter (mm)	Column Height (cm)	Column Volume (mL)
Desalting	26/400	16	85

DESALTING CHROMATOGRAM



✓ Ideal for desalting and buffer exchange of proteins and biomolecules

IDEAL APPLICATIONS



Molecular Weight Estimation



Aggregate / Monomer Analysis



Desalting & Buffer Exchange



Purity Assessment & Process Development



QC & Lot Release Testing

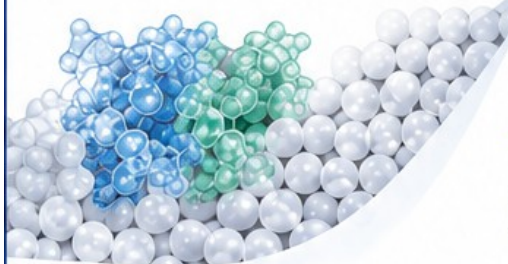
PROTPURE AGAROSE SEC RESIN



- ✓ Cross-linked Agarose Matrix
- ✓ High Physical & Chemical Stability
- ✓ Consistent Quality
- ✓ Scalable from Lab to Process

ProtPure

Purity. Performance. Reliability.



PROTPURE ION EXCHANGE RESINS

Agarose-Based Chromatography Media for
High-Performance Protein Purification

THREE ION-EXCHANGE MODES. THREE PERFORMANCE GRADES. ONE RELIABLE PLATFORM.

A highly cross-linked 6% agarose matrix engineered for excellent binding capacity, chemical stability and reproducible performance from **lab scale** to **manufacturing**.

DEAE AGAROSE



Weak Anion Exchanger

- Diethylaminoethyl (DEAE) ligand
- Ionic Capacity: 0.12 – 0.15 mmol Cl^-/mL resin
- Excellent for proteins with intermediate to basic pI
- Ideal for process development and intermediate purification

Applications



Protein purification & intermediate polishing



Enzyme & peptide purification



Vaccine & recombinant protein downstream processing



Separation of biomolecules with weak anionic nature

Q AGAROSE



Strong Anion Exchanger

- Quaternary Amine ligand
- Ionic Capacity: 0.18 – 0.25 mmol Cl^-/mL resin
- High selectivity for proteins with basic isoelectric points
- Excellent for capture, polishing and impurity removal

Applications



Monoclonal antibody & antibody fragment purification



Viral & plasmid DNA purification



Host-cell protein (HCP) removal



Process development & manufacturing-scale purification

SP AGAROSE



Strong Cation Exchanger

- Sulfopropyl (SP) ligand
- Ionic Capacity: 0.18 – 0.25 mmol H^+/mL resin
- High binding capacity with excellent selectivity
- Suitable for polishing and intermediate purification

Applications



Monoclonal antibody purification



Recombinant protein & enzyme purification



Vaccine & viral vector downstream processing



Intermediate purification & impurity removal

THREE PERFORMANCE GRADES TO MATCH YOUR PROCESS NEEDS



FAST FLOW

Higher throughput for higher productivity

Particle Size: 45 – 165 μm



PRECISE

Balanced performance for accuracy & reliability

Particle Size: 45 – 105 μm



HR (HIGH RESOLUTION)

Smaller particles for high-resolution separations

Particle Size: 25 – 55 μm



HIGH BINDING CAPACITY

High dynamic binding capacity for efficient capture and purification



BROAD pH STABILITY

CIP stability pH 2 – 14
Operational stability pH 2 – 12



ROBUST CHEMICAL STABILITY

Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine HCl and 70% ethanol



AGAROSE-BASED MATRIX

6% highly cross-linked spherical agarose for excellent performance



HIGH-RESOLUTION SEPARATIONS

Designed for superior selectivity and reproducibility



SCALABLE PERFORMANCE

From R&D and process development to manufacturing scale

IDEAL FOR



Protein Purification



Antibody Purification



Enzyme Purification



Vaccine & Viral Vector Downstream Processing



Plasmid & Viral DNA Purification



ENGINEERED FOR PURITY.
BUILT FOR PERFORMANCE.



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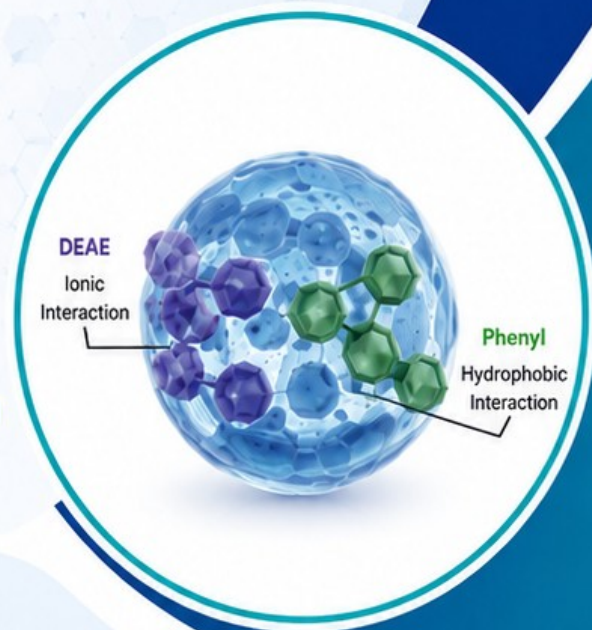


info@protpure.com

Hy-Ionic™ DP

DEAE-Phenyl Mixed-Mode Chromatography Resin

Dual Interaction. Enhanced Selectivity.



Hy-Ionic™ DP is a mixed-mode agarose chromatography resin combining DEAE and Phenyl functionalities on a single matrix, providing complementary ionic and hydrophobic interaction modes for enhanced chromatographic selectivity and process flexibility.

KEY FEATURES

- Dual functionality: DEAE (anion exchange) and Phenyl (hydrophobic interaction) on the same agarose matrix
- Two independently addressable interaction modes for flexible purification strategies
- High dynamic binding capacity in both modes
- Chemically stable – retains performance after prolonged alkaline exposure (0.1 M NaOH for 10 days)
- Enables mode switching on the same column without changing the resin
- Suitable for a wide range of biomolecules including antibodies, proteins and other biotherapeutics

TECHNICAL DETAILS

Matrix	Cross-linked Agarose
Functional Groups	DEAE and Phenyl
Mode of Interaction	Anion Exchange (DEAE) / Hydrophobic Interaction (Phenyl)
Particle Size	~ 60 – 150 µm (typical)
Application Range	Capture, Intermediate Purification and Polishing
pH Stability (Operating)	3 – 12 (typical)
Storage	20% Ethanol or as recommended

ORDERING INFORMATION

Product	Pack Size
Hy-Ionic™ DP (DEAE-Phenyl Mixed-Mode Resin)	5 mL, 25 mL, 100 mL, 1 L and custom packs

For bulk orders and custom packing, please contact us at info@protpure.com

DBC PERFORMANCE (10% BREAKTHROUGH)

DEAE MODE (Anion Exchange)	
Binding Buffer	20 mM Tris
Sample	BSA (5 mg/mL)
Elution Buffer	20 mM Tris + 1 M NaCl
Residence Time	4 min
DBC	100 mg BSA/mL resin

PHENYL MODE (Hydrophobic Interaction)	
Binding Buffer	2 M Ammonium Sulfate
Sample	BSA (5 mg/mL)
Elution Buffer	20 mM Tris
Residence Time	4 min
DBC	30 mg BSA/mL resin

CHEMICAL STABILITY

Same packed column (0.7 cm I.D. × 2.5 cm bed height) was stored in 0.1 M NaOH for 10 days. DBC experiments were repeated using the same column, and the results confirmed that the DBC was retained after 10-day storage in 0.1 M NaOH. **(Study continue)**

TYPICAL APPLICATIONS

- Antibody and Fc protein purification
- Recombinant protein purification
- Host Cell Protein (HCP) removal
- Plasmid DNA purification
- Biopharma process development

PHENYL AGAROSE

Hydrophobic Interaction Chromatography • HIC

High-performance HIC resin for protein purification with high capacity, stability and scalable flow performance.

BIOPROCESS RESIN
YES

Certificate of Analysis: Yes



PRODUCT OVERVIEW

Protpure Phenyl Agarose is a hydrophobic interaction chromatography resin based on a **6% cross-linked, spherical agarose matrix** with phenyl ligands. The resin is specified for protein and biomolecule purification applications requiring hydrophobic interaction selectivity, binding capacity, chemical stability and scalable flow performance.

PERFORMANCE AT A GLANCE



25–30
mg BSA/mL

DYNAMIC BINDING
CAPACITY



45–105
µm

PARTICLE SIZE



200–300
cm/h

FLOW PERFORMANCE
(<0.1 MPa)



3–13
pH

OPERATIONAL
pH RANGE

TECHNICAL SPECIFICATIONS

PARAMETER	SPECIFICATION
BioProcess resin	Yes
Certificate of Analysis	Yes
Ligand	Phenyl
Ligand concentration	~20–25 µmol phenyl/mL resin
Dynamic binding capacity	25–30 mg BSA/mL resin
Matrix	Cross-linked agarose, 6%, spherical
Particle size	45–105 µm
Particle size, d50V	~65 µm
Pressure / flow characteristics	200–300 cm/h at <0.1 MPa in an XK 50/60 column with 5 cm diameter and 25 cm bed height; at 20°C using buffers with the same viscosity as water.
pH stability, CIP	2–14
pH stability, operational	3–13
Storage	4–30°C, 20% ethanol

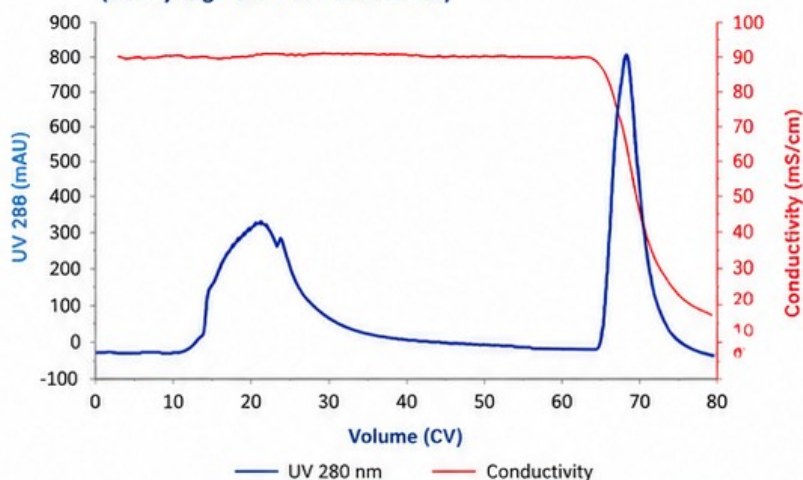


CHEMICAL STABILITY

Stable in commonly used aqueous buffers

- 1.0 M NaOH • 3 M ammonium sulphate
- 30% isopropanol • 70% ethanol
- 10% ethylene glycol • 0.5% SDS
- 6 M guanidine-hydrochloride • 8 M urea

TYPICAL PERFORMANCE (Phenyl Agarose HIC Purification)



High binding capacity and sharp elution profile for efficient protein purification using Phenyl Agarose HIC resin.



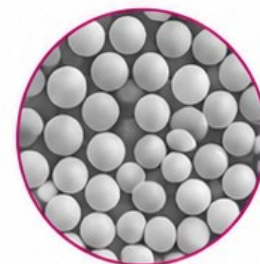
APPLICATIONS

- ✓ Protein purification and intermediate polishing
- ✓ Hydrophobic interaction chromatography (HIC)
- ✓ Process development and scale-up applications
- ✓ Purification of proteins and other biomolecules



KEY BENEFITS

- ✓ Phenyl ligand for hydrophobic interaction selectivity
- ✓ High BSA dynamic binding capacity
- ✓ Robust chemical stability
- ✓ Wide CIP and operational pH range
- ✓ BioProcess-ready resin platform



Evaluation samples & application support available upon request.

Contact us to discuss your process requirements.

Protpure Tech Private Limited

📍 A2, Plot No. A2/440/2, PNY, 425 Sq. Mtr.,
Opp. Paragon Paints, G.I.D.C., V.U. Nagar,
Anand – 388121, Gujarat, India

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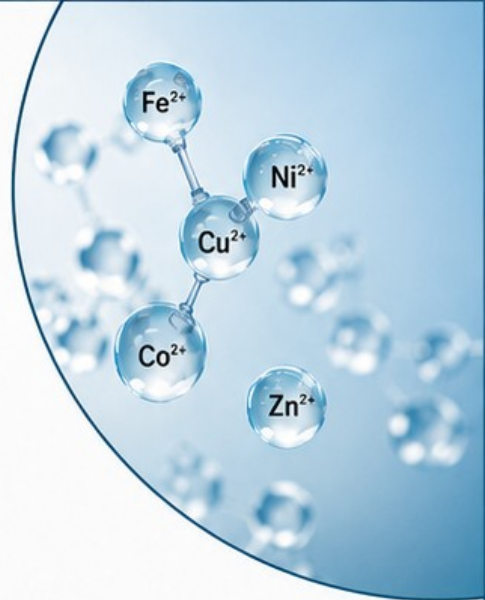
ProtPure MR Agarose

Transition Metal Removal Evaluation Kit

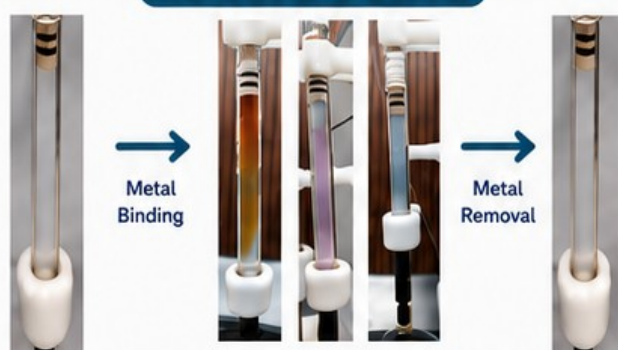
Validated metal removing technology
for bioprocessing applications



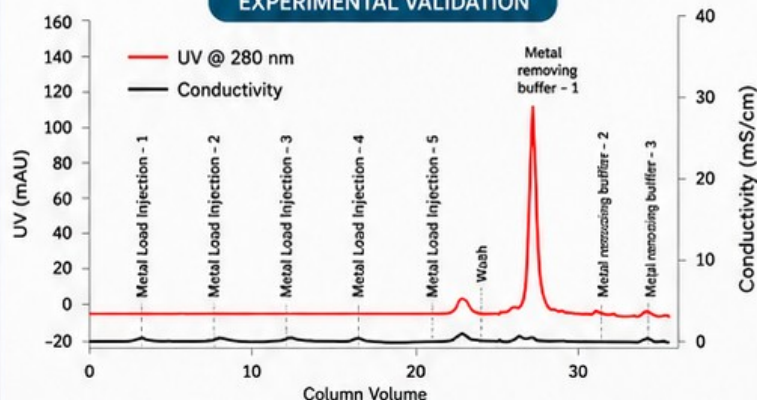
Metal binding performance was evaluated using protein purification FPLC system with sequential injection of transition metal ion salts under controlled experimental conditions.



METAL BINDING & REMOVAL



EXPERIMENTAL VALIDATION



**DYNAMIC METAL
BINDING CAPACITY**



**15–18 μmol transition metal ions
per 1 mL of MR Agarose.**

Validated for Fe+2 | Fe+3 | Ni+2 | Co+2 | Cu+2 | Zn+2



For more information:
www.protpure.com



Email:
info@protpure.com



Efficient removal of
transition metal ions



High binding capacity
& reproducibility



Easy to use protocol
for evaluation



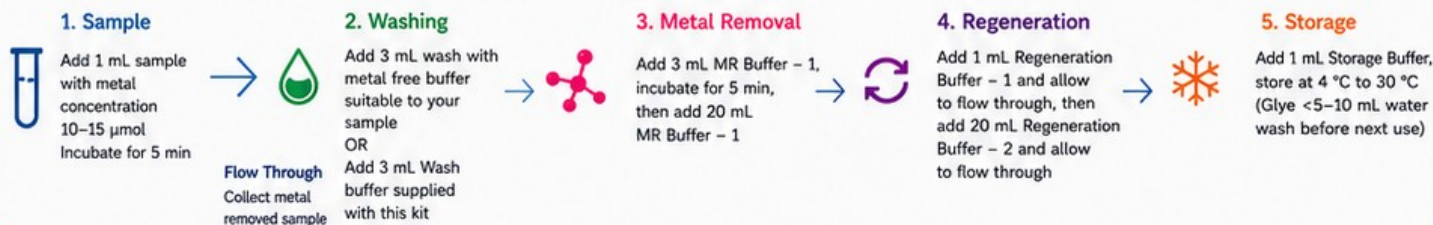
Ideal for rapid assessment
& method development

Kit Contents

Gravity Column with 1 mL MR Resin	1 Pc
Wash Buffer	30 mL
MR Buffer - 1	30 mL
MR Buffer - 2	200 mL
Regeneration Buffer - 1	10 mL
Regeneration Buffer - 2	200 mL
Storage Buffer	10 mL

Reaction No.: 10

EASY 5-STEP EVALUATION PROTOCOL



Note: Regeneration efficiency depends on metal species, metal concentration and operating conditions.

ProtPure

METAL AFFINITY CHROMATOGRAPHY

High-Performance Agarose-Based Resins for Efficient Purification of His-Tagged Proteins

Protpure's Metal Affinity Chromatography (IMAC) resins are designed for robust and selective purification of His-tagged proteins. Our agarose-based NTA resins deliver high binding capacity, excellent chemical stability, and consistent performance from research to large-scale processing.



HIGH BINDING CAPACITY

Efficient capture of His-tagged proteins.



FAST & EFFICIENT PURIFICATION

Faster processing with excellent flow properties.



CHEMICALLY STABLE

Agarose matrix ensures stability and reusability.



SCALE WITH CONFIDENCE

Consistent performance from R&D to production scale.

PROTPURE METAL AFFINITY RESINS

Ni-NTA Agarose

High-capacity Ni²⁺ affinity resin for general His-tagged protein purification.

Key Features

- ✓ High binding capacity
- ✓ Excellent flow properties
- ✓ Suitable for R&D to production scale
- ✓ Compatible with standard buffers
- ✓ Excellent chemical stability

Co-NTA Agarose

Co²⁺ affinity resin offering enhanced selectivity for specific His-tagged proteins.

Key Features

- ✓ Higher selectivity
- ✓ Reduced non-specific binding
- ✓ Ideal for challenging proteins
- ✓ Stable and reusable
- ✓ High chemical stability

Cu-NTA Agarose

Cu²⁺ affinity resin for strong binding and efficient purification.

Key Features

- ✓ High binding strength
- ✓ Effective for low-abundance proteins
- ✓ Fast binding kinetics
- ✓ Reliable performance
- ✓ Low metal leaching

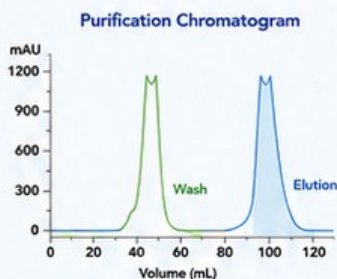
Zn-NTA Agarose

Zn²⁺ affinity resin with high chemical stability and broad compatibility.

Key Features

- ✓ High chemical stability
- ✓ Low metal leaching
- ✓ Broad pH and buffer compatibility
- ✓ Ideal for diverse applications
- ✓ Robust and reproducible

PROVEN PERFORMANCE



Efficient capture and elution of His-tagged protein using Protpure Ni-NTA Agarose.



SDS-PAGE analysis confirming high-purity His-tagged protein eluted from Protpure Ni-NTA Agarose.

KEY FEATURES (All Variants)

- ✓ **Agarose-Based Matrix** Provides high mechanical strength and excellent chemical stability.
- ✓ **High Binding Capacity** Optimized for efficient capture of His-tagged proteins.
- ✓ **Fast Flow Properties** Enables rapid processing with high productivity.
- ✓ **Wide Compatibility** Compatible with a range of buffers and conditions.
- ✓ **Excellent Reusability** Resins can be cleaned and reused for multiple cycles.
- ✓ **Consistent & Reproducible** Reliable performance batch after batch.

PRE-PACKED Ni-NTA COLUMNS

Ready-to-use columns packed with high-quality Ni-NTA Agarose for fast and convenient purification.



Consistent performance and reproducibility



Saves time and reduces setup



Available in multiple sizes for lab and pilot use



AVAILABLE COLUMN VOLUMES

1 mL

5 mL

10 mL

20 mL

TYPICAL APPLICATIONS



Recombinant protein purification



Protein expression screening



Antibody and fusion protein purification



Process development and scale-up



Research, biotech and biopharma workflows



ProtPure
ProtoPure Tech Pvt. Ltd.



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info@protpure.com

Chromatography solutions from resin screening to protein purification

01 PRECISION COLUMN PACKING

*Reproducible columns.
Reliable performance.*

Column Performance Report

- Asymmetry Factor (As)
- HETP
- Number of Theoretical Plates (N)
- Reduced Plate Height (h)

Column Diameters available

- 10 mm
- 16 mm
- 26 mm
- 50 mm



From 5 mL to 1 L volume

02 CHROMATOGRAPHY RESIN SCREENING & SELECTION

Identify the most suitable resin for your purification objective.



Binding



Recovery



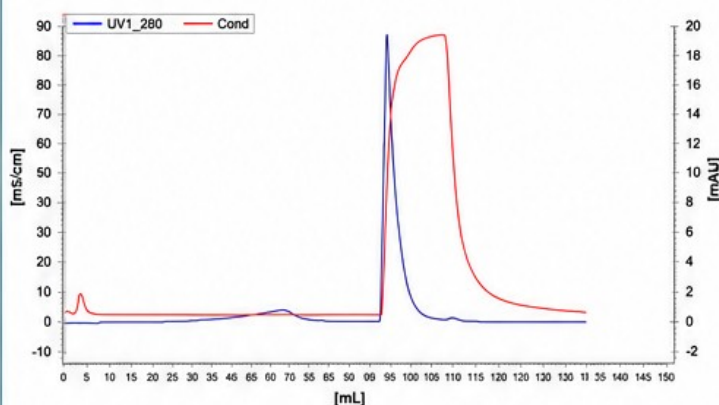
Selectivity



Resolution



Capacity

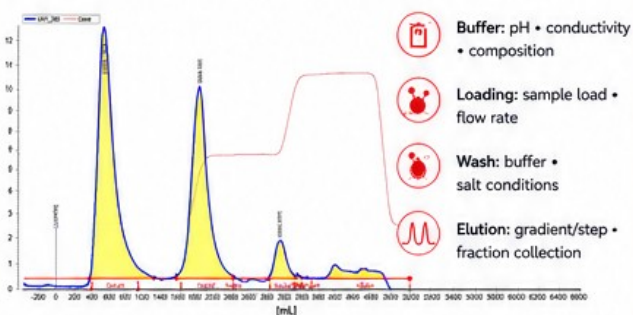


Screen multiple chromatography resins to identify the best purification route for your target protein.

03 METHOD DEVELOPMENT

Optimize chromatographic conditions for purity, recovery and reproducibility.

From sample to optimized chromatography method



Buffer: pH • conductivity • composition



Loading: sample load • flow rate



Wash: buffer • salt conditions



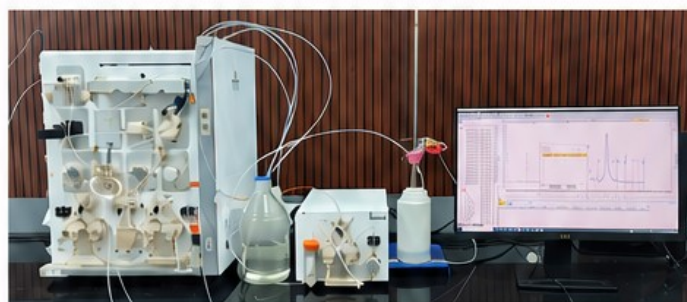
Elution: gradient/step • fraction collection



Optimization of buffer, loading, wash and elution conditions for improved purity and recovery.

04 PROTEIN PURIFICATION SERVICE

From clarified sample to purified protein.



Evaluation parameters:



DBC



Recovery



Selectivity



Pressure-flow



Resolution

PURIFICATION WORKFLOW



Clarification



Capture



Intermediate purification



Polishing



Analytical assessment

IDEAL FOR

- Academic & research laboratories
- Research centres
- CROs
- Biotech start-ups
- Industry



CHROMATOGRAPHY PLATFORMS

Affinity • Ion Exchange • HIC • SEC



We don't just purify your protein.
We select the resin, develop the method,
pack the column and demonstrate performance.